CPU Power Limits: Performance and Energy Efficiency Across Different Workloads

Group description

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The experiments will be run on Payam's desktop PC.

1. Introduction

Modern CPUs do not run at one fixed speed. They boost up and down depending on the workload, temperature and how much power they are allowed to use. More power usually means more performance, but probably not in a straight line. At some point the CPU may use quite a lot more power for only a small improvement.

That is what we want to test. We will set several CPU package power limits and run the same workloads at each limit. The main values we care about are performance, task time, average CPU package power, energy used for one completed task and temperature. Everything else should stay the same as far as possible.

Motivation and goal

Desktop CPU comparisons usually focus on the highest benchmark number. We think that misses part of the picture. If a computer loses only a few percent performance but uses much less energy and runs cooler, that may be a better setting for everyday use. It can also mean less fan noise, although we are not planning to measure noise in this project.

Our goal is to find out how the power/performance trade-off looks on one current AMD desktop CPU. We also want to see if the answer changes depending on the type of program. A Blender render can keep every core busy for a long time. A game behaves differently and may depend more on one or a few busy threads. Because of that, one “best” power limit may not exist for everything.

Research questions

- **RQ1:** How does the CPU power limit affect application performance?
- **RQ2:** Is the effect different for rendering, compression, video encoding and gaming?
- **RQ3:** How do performance per watt and the percentage performance gain relative to added CPU package energy change between two neighbouring PPT limits?
- **RQ4:** How much does the power limit change CPU temperature during longer workloads?

Objectives and hypotheses

We want to build a repeatable test procedure, collect the same set of measurements for every limit and compare the workloads. The stock setting will be our baseline.

Our main hypothesis is simple: when a workload is actually limited by the configured PPT, raising PPT should allow higher sustained clock speeds and improve performance. We expect measured package power and temperature to rise at the same time, while the performance gained per extra watt will fall at the higher limits. If a workload never reaches the limit, we expect little or no change. Rendering and encoding will probably react more strongly than gaming because they can load all CPU cores for longer. We also think at least one reduced limit will keep most of the stock performance while using less energy per task.

2. Related work

This general problem has already been studied, mostly with servers and research benchmarks. A common technique is dynamic voltage and frequency scaling (DVFS), where the processor changes its speed and voltage to balance performance and energy use.

Stachowski, Rünger and Rauber tested multi-threaded programs at different CPU frequencies [1]. One useful point from their work is that the fastest setting is not always the setting that uses the least energy. They also found that programs react differently. This is one reason we are not using only one benchmark.

Petoumenos et al. compared several ways of enforcing a CPU power cap [2]. Their results show that the capping method itself can affect performance and efficiency. Our project is narrower. We are not designing a new capping method or comparing algorithms. We will use one motherboard setting and focus on what happens to normal desktop workloads.

Raffin and Trystram looked at software-based CPU energy measurement on Intel and AMD systems [3]. They point out that these counters are useful, but they can also be misunderstood or measured badly. Because of this, we will use the same monitoring tool and sample rate in every run. We will call our result an estimate of CPU package energy, not exact wall-socket energy.

The selected related work largely uses server processors, Linux tools or benchmark suites such as PARSEC. Our planned test is different because it will use a recent Ryzen X3D desktop CPU, Windows and programs that a normal PC user might actually run. It is a smaller study, but it should still show whether the same performance-versus-energy trade-off appears on this system.

3. System

The system we plan to study includes the CPU, motherboard power settings, cooling, Windows, the benchmark programs and HWiNFO logging. The main independent variable will be PPT, the CPU package power limit. That setting affects the clock speed and voltage the CPU can hold under load. This can then change benchmark performance, measured package power and temperature.

The planned model is:

Power limit -> CPU boost behaviour -> performance, package power and temperature -> energy per task

For tasks with a clear start and finish, energy will be estimated with:

Energy (joules) ≈ sum of each logged package-power sample x its sampling interval

With a fixed logging interval, this is approximately the same as average logged package power multiplied by task time. This value is only an estimate of CPU package energy. It is not wall-socket energy or energy for the whole computer. For benchmarks that output a score, we will also calculate score per watt. For timed workloads, energy per task is more useful. A lower number means the CPU used less estimated package energy to finish exactly the same work. For RQ3, each step will report both the percentage performance change and percentage package-energy change instead of using a subjective description such as “small gain.”

The first planned limits are 65 W, 88 W, 105 W, 125 W, 150 W and stock/default. On this AMD platform the BIOS may call the limit PPT, or Package Power Tracking. We need to check what values the motherboard accepts before the real runs. The configured limit and the measured average power are not the same thing, so both will be recorded.

The stock/default baseline means that CPU power and boost settings are left at the motherboard's default or Auto values, with no manually entered PPT. The exact BIOS version and the final baseline values will be documented before testing. For the other configurations, we will keep the CPU at stock settings and change only PPT if the UEFI allows it.

4. Early environment and scope

Hardware and software

Part	Planned setup
CPU	AMD Ryzen 7 9850X3D, 8 cores/16 threads, 120 W default TDP [4]
Motherboard	ASRock X870E Taichi OCF
GPU	NVIDIA GeForce RTX 5080
Memory	V-Color DDR5, model TMXFL1680838KWK
Operating system	Windows 11 Pro
Monitoring	HWiNFO64 sensor logging
Power setting	Motherboard UEFI/PBO PPT control

The exact BIOS, chipset driver, GPU driver, memory speed and timings will be written down before testing. The same cooler, fan curve, power supply and case setup will be used for every run. AMD lists the 9850X3D as an 8-core, 16-thread Zen 5 processor with a 120 W default TDP [4]. The ASRock board supports Ryzen 9000 CPUs and has UEFI controls needed for the planned settings [5].

Workloads and measurements

We plan to use four workload types. Blender Benchmark in CPU mode will represent heavy multi-core rendering. The built-in 7-Zip benchmark will test compression and decompression. HandBrake will encode the same video file with the same CPU preset each time. We will also choose one game with a repeatable built-in benchmark.

Blender and HandBrake will mainly be measured with completion time. 7-Zip gives a rating, while the game will give average FPS and 1% low FPS. If it is feasible, an appropriate frame-capture tool such as PresentMon or CapFrameX will also be used to collect median, P95 and P99 frametime statistics. During every test we will log CPU package power, CPU temperature and effective clock. GPU use will also be checked during the game.

The game will run at low resolution and low graphics settings so the RTX 5080 is less likely to become the bottleneck. This may still not make the test completely CPU-limited. We are not fully sure which game will give the cleanest repeatable result yet, so choosing and checking that benchmark is part of the pilot stage.

This is a case study of one computer. We will not compare CPUs, operating systems, memory tuning, undervolting or different coolers. Whole-PC wall power is also outside the main scope unless we can get a reliable external meter. The results will therefore apply directly to this PC, not automatically to every Ryzen system.

Early method

First, we will save one BIOS profile for each limit and check that the setting really changes measured package power. PPT will be the independent variable. TDC, EDC, Curve Optimizer, boost override, the thermal protection and temperature limit, core count, memory configuration, GPU settings, fan curve and the Windows power plan will stay fixed. Stock CPU values will be used for these controls where possible. Any BIOS setting that must change in order to enter a manual PPT will be documented. Unneeded background programs will be closed.

Each workload will get one warm-up run. We will initially use at least five recorded repetitions for every workload and power limit. The pilot-test variability will be checked to decide whether more repetitions are needed. Power-limit configurations will be tested in a randomised or counterbalanced order to reduce temperature and time-order bias. Before a new run, the CPU should return to roughly the same idle temperature.

HWiNFO will use one fixed logging interval. We will keep the raw CSV logs, benchmark outputs, BIOS screenshots and a short experiment log. Results will mainly be shown as averages with standard deviation and 95% confidence intervals where they make sense. Graph axes will include names and units, and stock performance will be normalised to 100% for an easier comparison.

5. Early timeline

- **After A1 to 20 September:** use the A1 feedback, finish the reading and choose the exact benchmark versions and input files.
- **21-27 September:** prepare the BIOS profiles, confirm the controlled settings and run early pilot tests.
- **28 September:** present the experimental design for evaluation and note any required changes.
- **29 September-11 October:** revise the design, complete stability checks and finalise the logging procedure.
- **12 October-1 November:** run the main experiments and save the raw data and experiment logs.
- **2-8 November:** check data quality and repeat failed or clearly disturbed runs.
- **9-15 November:** calculate the metrics, create graphs and compare the workloads.
- **16-22 November:** revise the report and prepare the presentation.
- **Tentative 23 November:** present the project at the November conference and collect final feedback.

6. Discussion and risks

One risk is confusing the configured limit with real power use. A 105 W limit does not mean the CPU will use 105 W in every benchmark. It is a ceiling, not a target, so a lighter workload may stay well below it. We therefore need to record actual package power instead of using the BIOS number in the energy calculation.

Sensor accuracy is probably the biggest limitation. HWiNFO reports CPU package estimates, not total electricity from the wall. The values should still be useful for comparing runs if the setup stays fixed, but we cannot claim they are perfect physical measurements. Logging can also add a little overhead. We will keep its settings identical and avoid an unnecessarily high sample rate.

Temperature and background tasks are other risks. Ryzen boost changes with temperature, while Windows can start a service or update at a bad time. Repeated runs, a warm-up, mixed test order and saved logs should make unusual results easier to spot. We will not delete an outlier just because it looks inconvenient; there must be a reason in the experiment log.

There is also a safety and stability issue when changing BIOS settings. We will only change the package limit, leave temperature protection enabled and test each profile before collecting data. The memory setup must remain fixed and stable. Even after all of this, we may not find one obvious “best” limit. Rendering could prefer one point and gaming another. That would make the conclusion less neat, but it would also be a useful answer to RQ2.

References

[1] M. Stachowski, G. Rünger and T. Rauber, “Performance and energy metrics for multi-threaded applications on DVFS processors,” *Sustainable Computing: Informatics and Systems*, vol. 17, pp. 55-68, 2018. https://doi.org/10.1016/j.suscom.2017.10.015

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[3] G. Raffin and D. Trystram, “Dissecting the Software-Based Measurement of CPU Energy Consumption: A Comparative Analysis,” *IEEE Transactions on Parallel and Distributed Systems*, vol. 36, no. 1, pp. 96-107, 2025. https://doi.org/10.1109/TPDS.2024.3492336

[4] AMD, “AMD Ryzen 7 9850X3D Desktop Processor,” product specifications. https://www.amd.com/en/products/processors/desktops/ryzen/9000-series/amd-ryzen-7-9850x3d.html

[5] ASRock, “X870E Taichi OCF Specifications.” https://www.asrock.com/MB/AMD/X870E%20Taichi%20OCF/Specification.asp